

REQUEST FOR INFORMATION

AGENCY: U.S. Department of Homeland Security, Customs and Border Protection
Office of Field Operations, Non-Intrusive Inspection Program

TITLE: Artificial Intelligence for Image Adjudication

ISSUANCE DATE: April 30, 2026

THIS IS A REQUEST FOR INFORMATION (RFI) ONLY. This RFI is issued solely for information and planning purposes; it does not constitute a Request for Proposal (RFP) nor a promise to issue an RFP or other solicitation. Furthermore, those who respond to this RFI should not anticipate feedback about its submission, other than acknowledgment of receipt. This RFI does not commit the Government to contract for any supply or service. The Department of Homeland Security (DHS) Customs and Border Protection (CBP) is not seeking proposals at this time. Respondents are advised that the U.S. Government will not pay any cost incurred in response to this RFI. All costs associated with responding to this RFI will be solely at the expense of the respondent. Not responding to this RFI does not preclude participation in any future solicitation. The information provided in this RFI is subject to change and is not binding on the Government. All submissions become the property of the Federal Government and will not be returned.

1.0 Background

CBP utilizes a suite of technology, techniques, and tactics as part of its multi-layered approach to meet the agency's enduring mission priorities. Non-Intrusive Inspection (NII) systems are an integral part of CBP multi-layered enforcement strategy to assist personnel in examining cargo, baggage, and conveyances for illicit merchandise and substances without physically opening or unloading them. NII systems allow CBP to detect anomalies and materials that could pose potential threats to the U.S. economy and national security, while maintaining the flow of lawful trade and travel. NII systems include, but are not limited to, low and multi-energy X-ray portals (LEPs and MEPs), radiation portal monitors (RPMs), and chemical detectors. NII technology is currently deployed to airports, seaports, international mail facilities, express consignment, rail crossings, and land border ports of entry (POEs).

For this RFI, CBP is specifically focused on large-scale (LS) imaging systems used to scan privately owned vehicles (POVs) and cargo vehicles (e.g. commercial trucks and similar conveyances). These systems generate high volumes of X-ray image data and associated metadata. Many large-scale image systems support the World Customs Organization (WCO) Unified File Format (UFF) as the accepted standard for packaging and sharing image data and metadata. CBP recognizes, however, that not all historical or legacy data will conform to WCO UFF, and that metadata fields may be incomplete, inconsistent, or inaccurate across locations, times, and OEMs.

CBP is expanding the coverage of NII scanning technology with modern, integrated systems capable of greatly increasing scanning throughput. The sheer volume of data generated by increased scanning

can be overwhelming for human operators, necessitating innovative solutions to maintain efficiency and effectiveness. Artificial Intelligence (AI) and Machine Learning (ML) models can be leveraged as tools to assist CBP employees with making better decisions faster, reviewing and synthesizing data, improving efficiency and learning from every new piece of information. Over the next few years, CBP plans to incorporate AI/ML tools into NII and other CBP processes, explore and test AI/ML use cases, and monitor advancements to AI/ML tools worldwide.

CBP is seeking AI/ML tools to assist in NII image adjudication, to include anomaly detection, using a human-in-the-loop approach to ensure model integrity. It is important to note that CBP's frontline personnel remain the most critical asset to thwarting illicit goods crossing our borders. AI/ML serves as a tool. A key goal of AI/ML use is to reduce the average NII examination processing time to achieve: (1) an increase in the number of cargo conveyances, private vehicles, packages, containers, and individuals scanned, (2) improved security, and (3) a reduction in the number of officer / agent hours used to conduct NII examinations – all without impact to the flow of legitimate trade and travel.

While CBP's most valued resource is its employees, CBP recognizes that technology is a vital element to mission success. CBP believes AI/ML tools will revolutionize border security and the flow of trade by seamlessly integrating NII systems and processes with relevant algorithms and other tools that will enable proactive threat detection, optimize resource allocation and ensure the swift and secure flow of commerce.

2.0 AI/ML Goals and Objectives

CBP seeks to incrementally implement AI/ML models that provide computer-assisted analysis of NII images and other data to increase the effectiveness and efficiency of NII operations. The AI/ML goals and objectives include:

1. Incrementally implement AI/ML models that provide computer-assisted analysis of NII images and other data to increase the effectiveness and efficiency of NII operations, with an emphasis on very high daily volumes of LS-NII scans of POVs and cargo vehicles.
2. CBP is not seeking proposals whose primary deliverable is a proprietary data platform, data pipeline, or case management system. CBP anticipates using existing and planned CBP-owned data platforms, pipelines, and viewers, and seeks algorithmic components that can be integrated into that environment.
3. Determine if AI/ML models can provide automated identification of anomalies, identification of conveyance modifications, manifest validation, contraband and narcotic identification, identification/verification of commodities or some combination of these capabilities through a series of algorithms applied to LS NII imagery and metadata.
4. Use AI/ML models to assist officer image analysis on higher-risk shipments, identify higher-risk areas of interest, and support a cost-effective, efficient operation.

CBP recognizes that human performance is difficult to quantify, varies by context, and may not lend itself to a single benchmark. Metrics and evaluation approaches will need to be tailored to specific algorithms, roles, and operational needs.

3.0 Purpose of this RFI

Through this RFI, CBP is seeking information from industry on how to best develop AI/ML models in concert with existing integrated NII system deployments, implementation of new X-ray scanning concepts of operation (CONOPs), and in response to current and anticipated future threats.

CBP is particularly interested in modular algorithms and models that:

- Operate on large scale NII imagery and associated metadata from POV and cargo vehicle scanners (with emphasis on WCO UFF where available);
- Can be adapted to CBP operational needs and constraints; and
- Can be integrated as drop-in components within existing CBP architectures, rather than requiring CBP to acquire or license new proprietary data platforms or data pipelines.

CBP does not wish to constrain developers' imaginations with a fixed set of use cases. The examples below are illustrative. At the same time, CBP must temper expectations regarding uniformity, completeness, and reliability of metadata, and the ability to define universal human performance baselines

Vendors should address the following questions in their RFI response (tailored to large scale NII POV and cargo vehicle imagery and associated metadata):

1. Describe algorithm products that you have developed, or are developing, for the following or related applications, including associated performance metrics:
 - a. Anomaly detection
 - b. Manifest / commodity verification
 - c. Conveyance modification detection (concealment locations, etc.)
 - d. Contraband detection (narcotics, weapons, currency, organic powders vs. organic goods)
2. Describe your development approach and any key limitations, including:
 - a. Types of models/architectures used and how they generalize across different ports, sensors, and operating conditions
 - b. Known limitations, sensitivities, or failure modes relevant to LS-NII POV and cargo vehicle imagery
3. Describe your capabilities on breaking down conveyance into components regions (segmentation). How does this support the applications described above?
4. Is your algorithm image-type dependent (jpg, tiff, png, proprietary LS-NII X-ray formats, etc.)? Please describe supported image formats, including any current or planned support for UFF and how your solution ingests and uses UFF image data and metadata.
5. Are there technical dependencies with your algorithm (system requirements)? Please describe require or recommended hardware, operating systems, and software dependencies for deployment at scale in CBP environments, particularly in cloud-based architectures.

6. Are there any technical incompatibilities? Particularly with industry accepted technologies (e.g., Windows vs Linux, Java vs .net, etc.). Please explain.
7. What does a typical architecture look like? In other words, what key technologies are required, and how would your solution be trained, deployed, and scaled to handle tens of thousands of LS-NII images per day across multiple scanner types.
8. Have your products been field tested:
 - a. If so, what were the operational settings (vehicle types, environments, throughput)?
 - b. What is the scope of commodities and modalities fielded and where?
 - c. What is the average volume handled by your solution, in terms of number of images processed daily and size of the average image?
 - d. What is the Technology Readiness Level?
 - e. What are the performance results, including detection and false alarm rates where available?
9. Describe your methods and requirements for training, verification, and deployment:
 - a. Do you have your own test data or is government data required? If government data is required, what metadata would be required, and what is the minimum metadata you can effectively use?
 - b. How does your solution perform when metadata is incomplete, inconsistent, or inaccurate even provided in the correct format?
 - c. Access to CBP systems requires a Background Investigation (BI). Do you or any of your employees have a CBP BI? If not, would you be able to test or train at a CBP location?
 - d. What internal company sources of NII data can be leveraged or made available to the government for stream of commerce, threat images, simulated threat images?
 - e. Describe how you deal with or have dealt with limited or absent labeling or metadata when training and validating your models.
10. Describe your algorithm system integration approach, including:
 - a. File formats and APIs?
 - b. Common Integration Platform / Common Viewer integration?
 - c. If you have a stand-alone and/or proprietary viewer, what steps can you take to integrate or consolidate with CBP's viewer so that CBP can use its existing tools while still leveraging your algorithms?
 - d. Are the AI/ML models compatible, or can they be made compatible, with algorithms generated by other parties in parallel or cascading operation (e.g., standard data structures or schemas for exchanging detections and scores)?
11. Describe how areas of interest, risk scores, and other model results are displayed to officers and how officer feedback is captured, if applicable.
12. How is your AI/ML engine evolving?
 - a. Does the AI/ML model evolve through continuous training? If so, please explain your approach to continuous or periodic updates, including governance and safeguards.

- b. What is your approach to monitoring performance over time and detecting model drift or degradation?
13. With respect to the applications above, please indicate:
 - a. Which types of contraband or conditions (e.g., narcotics, weapons, generic anomalies, modifications, currency, other prohibited or high-risk items) your AI/ML currently addresses or has been trained on?
 - b. Which LS-NII modalities are supported (e.g., POVs, cargo vehicles, sea containers, rail cargo), and what would be required to extend to additional conveyance types if needed?
14. What investment, acquisition, design, and management strategies would be conducive to minimizing technical risk and life cycle costs while accelerating development and deployment of algorithms into CBP's existing LS-NII architectures?
15. How can industry help to standardize and leverage technology and software solutions for use across various ports of entry and other facilities to maximize interoperability across systems, when feasible, while still allowing CBP to rely on its own data platforms and pipelines?
16. What subcontract / teaming arrangements, including small / disadvantaged businesses, would you put in place to address the full range of stated requirements (algorithm development, integration, sustainment, operator support)?
17. What collaborative activities should government and private sector entities undertake to encourage the development and adoption of open architecture, non-proprietary AI/ML solutions that can be integrated into diverse LS-NII environments?

4.0 RFI Response Instructions

Only electronic submissions will be accepted. Electronic files larger than 10 MB shall be broken down into multiple files with no one file exceeding 10 MB. Each electronic submission shall include:

1. The RFI title in the subject line
2. The maximum number of pages per submission shall be limited to fifteen (15) pages excluding title page, table of contents, company information page and reference materials, if attached.
3. All content must be provided in Microsoft Word or PDF format.
4. The company information page shall include the following information:
 - Company Name
 - Socio-economic status
 - DUNS number, UEI number, and CAGE code
 - Point-of-Contact information for technical questions
 - Point-of-Contact information for contractual questions
 - Federal Contract(s) available for use
5. The file number and total files being submitted (e.g., File 1 of 4, File 2 of 4, etc.) in the

subject line, if submission includes more than one electronic email.

This notice is part of Government Market Research, a continuous process for obtaining the latest information on the commercial status of the industry. This RFI is for planning purposes, and does not constitute a commitment, implied or otherwise, that a procurement action will follow.

CBP will use the information submitted in response to this notice at its discretion and will not provide comments to any submission; however, CBP reserves the right to contact any respondent to this notice for the sole purpose of enhancing CBP's understanding of the notice submission. The content of any responses to this notice may affect any subsequent solicitation.

Responses to this notice are not offers and cannot be accepted by the Government to form a binding contract. Responders are solely responsible for all expenses associated with any voluntary response to this RFI, including any subsequent product demonstrations.

Questions and responses on this RFI must be sent via email to the following:

Aga Frys, Contracting Officer – aga.frys@cbp.dhs.gov

Max Hempe, Contracting Specialist – max.hempe@cbp.dhs.gov

Questions must be submitted by 12:00 PM EST, May 15, 2026. Include the RFI title in the subject line of your email when submitting questions.

Responses to this RFI must be sent via email to the email address above no later than 12:00PM EST, May 30, 2026.